

Adaptive Advance-Time Control for nFAPI Split under Unpredictable Latency

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Abstract—

Index Terms—nFAPI, O-RAN, Functional Split, Adaptive Control.

I. Introduction

A. Background and Motivation

With the evolution of 5G and the Open Radio Access Network (O-RAN), architectures based on functional splits have become the key to improving network flexibility and reducing deployment costs. Currently, the most mainstream implementation promoted by the O-RAN Alliance is Split Option 7.2 (Split 7.2). Because Split 7.2 divides the Physical (PHY) layer into high and low parts and transmits continuous in-phase and quadrature (IQ) samples, it imposes extremely high requirements on the fronthaul network.

Compared to Split 7.2, the 3rd Generation Partnership Project (3GPP) Split Option 6 (Split 6), which separates the network at the Media Access Control (MAC) and PHY layers, offers a more flexible alternative. First, Split 7.2 relies on hardware-based synchronization protocols, such as IEEE 1588 Precision Time Protocol (PTP) and Synchronous Ethernet (SyncE), to achieve nanosecond-level phase and time synchronization. Its timing window requirement is typically strict, often needing to be less than 100 μ s. In contrast, the nFAPI (Network Functional Application Platform Interface) protocol, which is a prominent implementation of Split 6, employs a message-based synchronization mechanism. Through its proprietary P7 Node Sync procedure, nFAPI achieves slot-level synchronization (millisecond-level) and provides a highly flexible timing window ranging from 0 to 30,000 μ s. Furthermore, in terms of payload, Split 6 transports packetized MAC Protocol Data Units (PDUs) rather than continuous IQ samples. These characteristics allow Split 6 to demonstrate better adaptability in non-ideal fronthaul network environments where high jitter from operating system (OS) scheduling and network delivery is present.

This paper chooses Split 6 as the primary research subject due to its significant deployment benefits. As indicated by Pérez-Romero et al. [1], Split 6 effectively reduces the traffic burden on the fronthaul network. Compared to low-layer split schemes that transmit frequency-domain signals, its bandwidth requirement is significantly lower. In addition, Khan et al. [2] demonstrate that for the deployment of sub-6 GHz small cell networks, Split 6 can maintain robust transmission

performance over non-ideal fronthaul connections, thereby achieving an optimal cost-revenue trade-off.

B. Related Works

C. Contributions

The main contributions of this paper are summarized as follows.

- Contribution 1.
- Contribution 2.
- Contribution 3.

The rest of the paper is organized as follows. Section II defines the system model/architecture considered in this paper. The proposed analytical model/method is given in Section III. Section IV shows the numerical results. Concluding remarks are given in Section VI.

II. System Model/Architecture

In recent years, the Open Radio Access Network (O-RAN) architecture has become popular. O-RAN allows for more flexible network deployments. The Network Functional Application Platform Interface (nFAPI) is an important part of this architecture. nFAPI helps split the O-RAN Distributed Unit (O-DU) into O-DU High and O-DU Low parts. According to the nFAPI specification [3], the Virtual Network Function (VNF) includes Layer 2 and Layer 3 operations. The Physical Network Function (PNF) handles the Layer 1 Physical layer processing. This functional split physically separates the Media Access Control (MAC) layer and the Physical (PHY) layer. Crucially, the MAC scheduler responsible for allocating radio resources resides within the VNF. Figure 1 shows this O-RAN nFAPI functional split architecture and the separation of these components.

Deploying the VNF and PNF on different machines introduces significant challenges. Because the MAC scheduler is separated from the PHY layer, strict slot synchronization is required between these distributed components. However, even when deployed in the exact same network environment, different traffic loads directly impact the latency. Figure 2 illustrates the latency between the VNF scheduler and the PNF under varying traffic loads. As traffic increases, three factors heighten the overall delay: the scheduler requires more processing time, the data packing time lengthens, and the VNF-to-PNF One-Way Delay (OWD) is affected. Consequently, the VNF cannot reliably transmit data exactly

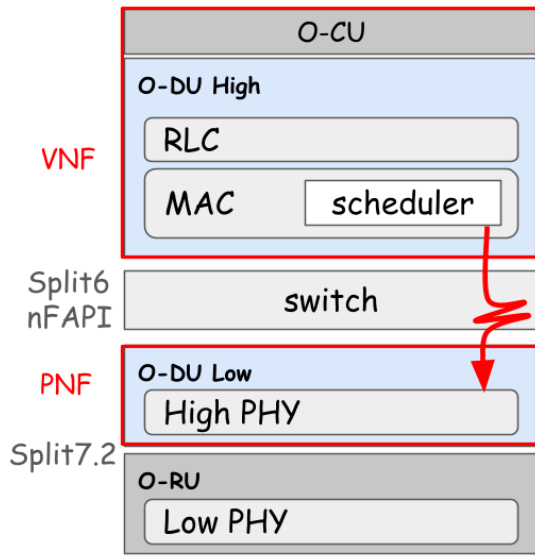


Fig. 1. O-RAN nFAPI functional split architecture.

at the slot boundary. Instead, it must dynamically advance its scheduling point to compensate for these varying delays and ensure packets arrive at the PNF on time.

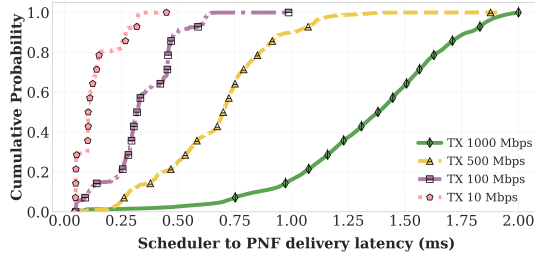


Fig. 2. Observation of latency between VNF scheduler and PNF under different traffic loads.

To lower deployment costs, operators often want to deploy Split-6 nFAPI gNBs over non-ideal networks. However, beyond traffic load variations, these non-ideal networks introduce additional, unpredictable latency. Figure 3 shows the latency jitter caused by intermediate routers and switches in a non-ideal deployment. Existing open-source solutions, such as OpenAirInterface (OAI), provide basic nFAPI support but were not designed to fully comply with the nFAPI specification's delay management procedures [3]. Therefore, they suffer from synchronization failures when encountering unpredictable latency from routers and switches. To address these challenges, this paper proposes a dynamic trigger scheduler adjustment algorithm that fully complies with the specification. Through systematic experiments, we verify the effectiveness of this dynamic adjustment and its benefits to overall network reliability.

The following assumptions were made in this paper.

- Assumption 1.
- Assumption 2.

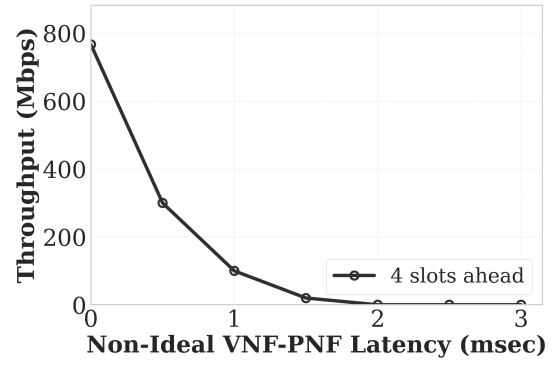


Fig. 3. Impact of unpredictable network latency in a non-ideal environment.

TABLE I
nFAPI Delay Management Configuration Parameters

Parameter	Type	Description
timingWindow	uint16_t	Timing window for delay management. See section 3.2.9 of [10]
timingMode	uint8_t	Timing mode for Timing Info. See section 3.2.9 of [10]
timingPeriod	uint8_t	Timing period for Timing Info. See section 3.2.9 of [10]

- Assumption 3.

III. Proposed Method

IV. Numerical/Simulation/Experimental Results

Computer simulations/Experiments were conducted to verify xxx (the effectiveness of the proposed analytical model). In the following figures, each point represented the average value of 10^5 samples. In each sample, we collect the statistics of xxx. Symbols and lines were used to show the simulation and analytical results, respectively.

XXX scenarios were investigated. Scenario I was designed to demonstrate/verify the accuracy/effectiveness of Contribution 1.

A. Contribution 1

Fig. 1.1: Purpose X axis: Y axis:

B. Contribution 2

Fig. 2.1: X axis: Y axis:

C. Contribution 3

Fig. 3.1: X axis: Y axis:

References

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